

DDL Observation

Tables used: STUDENT, FACULTY, DEPARTMENT/DEPT, COURSE.

-- 1. CREATE: Create database and table

```
CREATE DATABASE company;
USE company;

CREATE TABLE employees (
  emp_id INT PRIMARY KEY AUTO_INCREMENT,
  emp_name VARCHAR(50) NOT NULL,
  department VARCHAR(30),
  salary DECIMAL(10,2),
  hire_date DATE
);
```

```
mysql> CREATE TABLE employees (
->   emp_id INT PRIMARY KEY AUTO_INCREMENT,
->   emp_name VARCHAR(50) NOT NULL,
->   department VARCHAR(30),
->   salary DECIMAL(10,2),
->   hire_date DATE
-> );
```

Query OK, 0 rows affected (0.06 sec)

```
mysql> desc employees;
```

Field	Type	Null	Key	Default	Extra
emp_id	int	NO	PRI	NULL	auto_increment
emp_name	varchar(50)	NO		NULL	
department	varchar(30)	YES		NULL	
salary	decimal(10,2)	YES		NULL	
hire_date	date	YES		NULL	

5 rows in set (0.03 sec)

-- 2. ALTER TABLE: Add columns

```
-- Modify FACULTY by adding column Dept VARCHAR(10)
ALTER TABLE FACULTY ADD Dept VARCHAR(10);
ALTER TABLE FACULTY ADD Age NUMBER(3);  -- fails: NUMBER is not a valid MySQL
type
```

```
-- ALTER TABLE - DROP COLUMN (demonstration)
ALTER TABLE STUDENT ADD TempCol VARCHAR(5);  -- add a throwaway column
ALTER TABLE STUDENT DROP COLUMN TempCol;      -- then drop it
```

```
mysql> ALTER TABLE FACULTY ADD Dept VARCHAR(10);
Query OK, 0 rows affected (0.10 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE FACULTY ADD Age NUMBER(3);
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that
  ght syntax to use near 'NUMBER(3)' at line 1
mysql> ALTER TABLE STUDENT ADD TempCol VARCHAR(5);
Query OK, 0 rows affected (0.08 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE STUDENT DROP COLUMN TempCol;
Query OK, 0 rows affected (0.07 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc faculty;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| FacNo | varchar(4) | YES | | NULL | |
| FacName | varchar(15) | YES | | NULL | |
| Gender | char(1) | YES | | NULL | |
| DOB | date | YES | | NULL | |
| DOJ | date | YES | | NULL | |
| MobileNo | bigint | YES | | NULL | |
| Dept | varchar(10) | YES | | NULL | |
+-----+-----+-----+-----+-----+
7 rows in set (0.01 sec)
```

-- 3. Adding Constraint: PRIMARY KEY

```
ALTER TABLE STUDENT    ADD PRIMARY KEY (RegNo);
ALTER TABLE FACULTY    ADD PRIMARY KEY (FacNo);
ALTER TABLE DEPARTMENT ADD PRIMARY KEY (DeptNo);
ALTER TABLE COURSE     ADD PRIMARY KEY (CourseNo);
```

```
mysql> ALTER TABLE STUDENT ADD PRIMARY KEY (RegNo);
Query OK, 0 rows affected (0.07 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> ALTER TABLE FACULTY ADD PRIMARY KEY (FacNo);
Query OK, 0 rows affected (0.04 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> ALTER TABLE DEPARTMENT ADD PRIMARY KEY (DeptNo);
Query OK, 0 rows affected (0.05 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> ALTER TABLE COURSE ADD PRIMARY KEY (CourseNo);
Query OK, 0 rows affected (0.04 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> desc course;
```

Field	Type	Null	Key	Default	Extra
CourseNo	varchar(3)	NO	PRI	NULL	
CourseDesc	varchar(14)	YES		NULL	
CourseType	char(1)	YES		NULL	
SemNo	char(1)	YES		NULL	
HallNo	varchar(4)	YES		NULL	
FacNo	varchar(4)	YES		NULL	

-- 4. ALTER TABLE - ADD FOREIGN KEY constraint

```
ALTER TABLE COURSE ADD CONSTRAINT fk_course_faculty
FOREIGN KEY (FacNo) REFERENCES FACULTY(FacNo);
```

```
ALTER TABLE DEPARTMENT ADD CONSTRAINT fk_dept_head
FOREIGN KEY (DeptHead) REFERENCES FACULTY(FacNo);
```

```
mysql> ALTER TABLE COURSE ADD CONSTRAINT fk_course_faculty
-> FOREIGN KEY (FacNo) REFERENCES FACULTY(FacNo);
Query OK, 0 rows affected (0.12 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql>
mysql> ALTER TABLE DEPARTMENT ADD CONSTRAINT fk_dept_head
-> FOREIGN KEY (DeptHead) REFERENCES FACULTY(FacNo);
Query OK, 0 rows affected (0.10 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

-- 5. RENAME TABLE

```
RENAME TABLE DEPARTMENT TO DEPT;
```

```
mysql> RENAME TABLE DEPARTMENT TO DEPT;
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> desc dept;
```

Field	Type	Null	Key	Default	Extra
DeptNo	varchar(4)	NO	PRI	NULL	
DeptName	varchar(15)	YES		NULL	
DeptHead	varchar(4)	YES	MUL	NULL	

3 rows in set (0.06 sec)

-- 6. TRUNCATE TABLE and DROP TABLE

```
CREATE TABLE SCRATCH (id INT);
INSERT INTO SCRATCH VALUES (1), (2), (3);
TRUNCATE TABLE SCRATCH;
DROP TABLE SCRATCH;
```

```
mysql> CREATE TABLE SCRATCH (id INT);
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO SCRATCH VALUES (1), (2), (3);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> TRUNCATE TABLE SCRATCH;
Query OK, 0 rows affected (0.04 sec)

mysql> drop table scratch;
Query OK, 0 rows affected (0.04 sec)
```